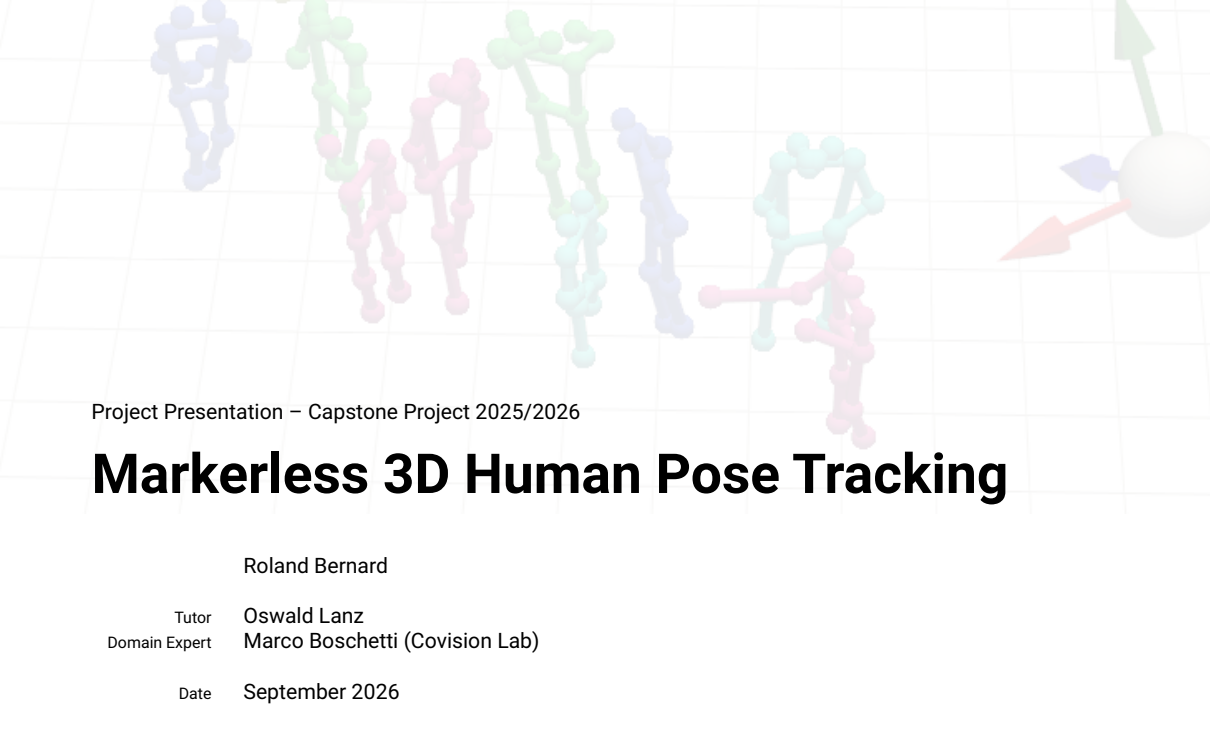




Project Presentation – Capstone Project 2025/2026

Markerless 3D Human Pose Tracking

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Introduction

- ▶ Problem: Track poses of multiple people in videos across multiple views.
- ▶ Possible Applications:
 - Robotics and human-robot interactions.
 - In healthcare, for tasks such as gait analysis or remote patient monitoring.
 - For real-time motion capture or natural body-based control interfaces for gaming.
- ▶ Objectives of this project:
 - Develop and evaluate a 3D pose tracking pipeline.



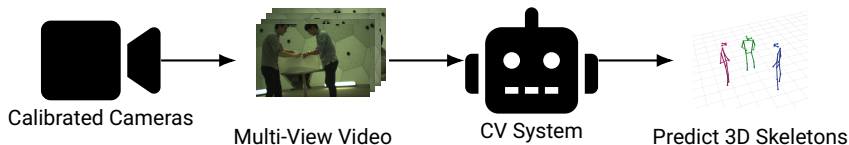
Task Formulation

- ▶ Given:

- Multiple video sequences containing multiple people.
- Recorded synchronously from multiple camera views.
- Camera calibration parameters, including extrinsics, intrinsic and distortion.

- ▶ Output:

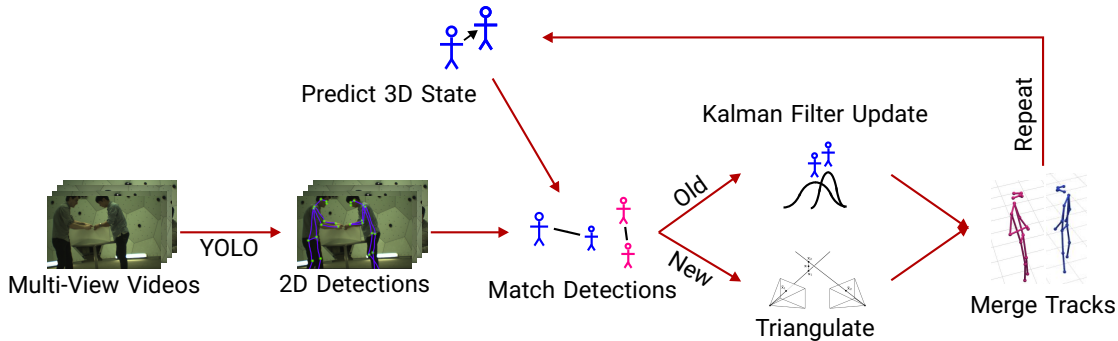
- 3D coordinates of skeletal joints for each subject.
- Temporally consistent movement of skeletal joints over time.



System Architecture – Overview

► Summary:

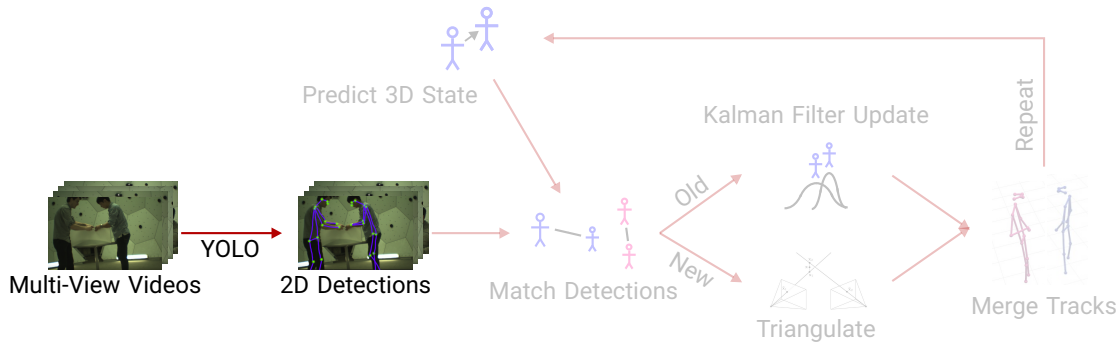
- Perform single-view single-frame 2D estimation using pretrained model.
- Cross-view and prediction-view matching of detections using Mahalanobis distance.
- Fusion of 2D observations into 3D skeletal trajectories using extended Kalman filter.



System Architecture – 2D Detections

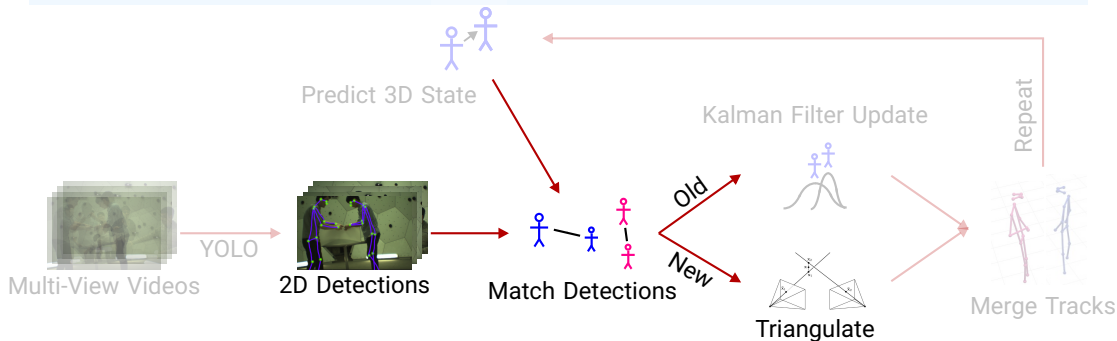
► Summary:

- Using YOLO26-pose family of models for the detector.
- Dynamically scale observation covariances based on YOLO keypoint confidence.
- Transform 2D pixel coordinates into normalized camera coordinates.



System Architecture – Data Association

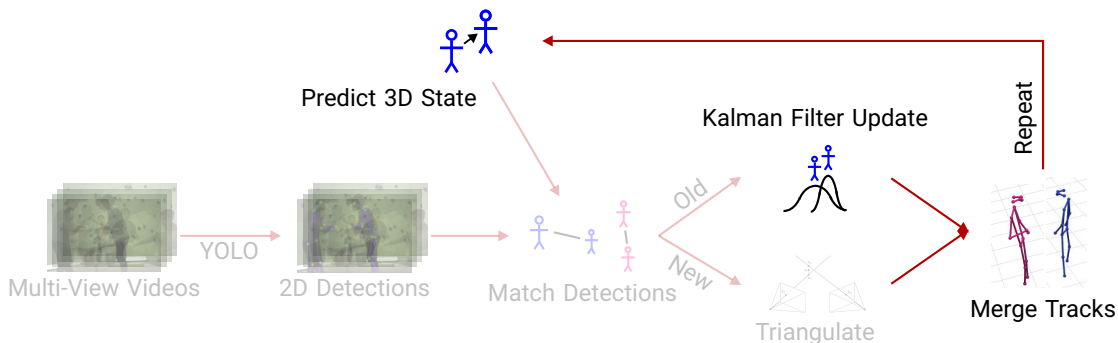
- ▶ Two-step process:
 - For each view, try to associate detections to predicted 3D state.
 - For all unmatched detections, try to associate between different views.
- ▶ For all cross-view matches:
 - Triangulate 3D keypoint locations using all matched views to initialize tracks.



System Architecture – Extended Kalman Filtering

► Summary:

- State representation contains 3D position and velocity for each keypoint.
- Prediction uses standard linear velocity model.
- All associated observations are concatenated and considered at once.
- Since the observation process is non-linear extended Kalman filtering is used.



Skeletal Constraints

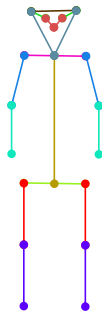
► Summary:

- Assume limb lengths do not change and are left-right symmetrical.
- Adding length values to the state vector of the filter.
- No direct interaction with other state values and very low process noise.
- Pseudo-observation merged into each update step that observes constraint violation of 0.

$$\mathbf{x} = [\mathbf{p}_1^T, \dots, \mathbf{p}_{17}^T, \mathbf{v}_1^T, \dots, \mathbf{v}_{17}^T, l_1, \dots, l_{12}]^T$$

$$h_{\text{sc}}(\mathbf{x}) = [\rho_1(\mathbf{x}) \quad \rho_2(\mathbf{x}) \quad \dots \quad \rho_{20}(\mathbf{p})]^T$$

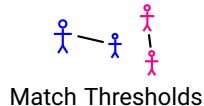
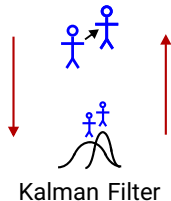
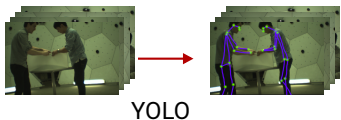
$$\rho_j(\mathbf{x}) = \|\mathbf{p}_{a_j} - \mathbf{p}_{b_j}\|_2 - l_{ij} \quad \mathbf{z}_{\text{sc}} = 0$$



Learning Parameters from Data

► Summary:

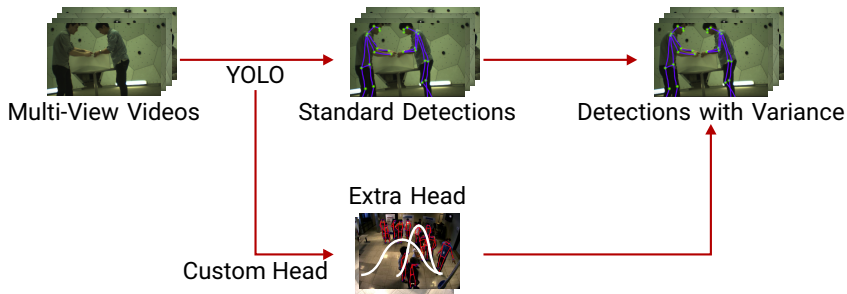
- Learning the complete pipeline end-to-end unfeasible.
- Split the learning into three independent parts that are learned separately.
- Training is done using the CMU Panoptic dataset excluding sequences used for testing.
- Sequences reserved for testing are excluded from the training.



Learning Parameters from Data – Custom YOLO Head

► Summary:

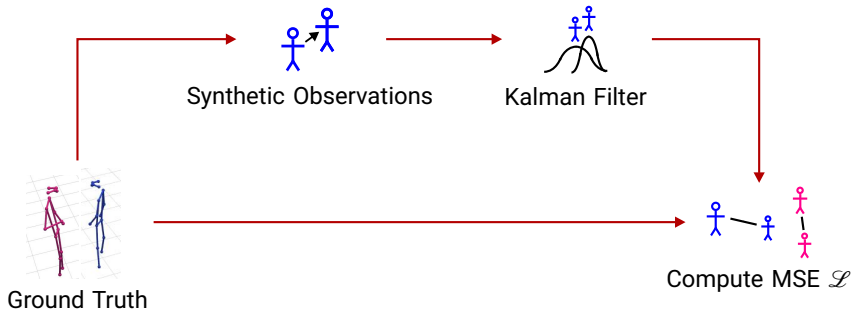
- Standard YOLO does not provide uncertainty output and does not predict invisible joints.
- Dataset generated from CMU Panoptic by training random camera view every second.
- Custom head outputs x and y offsets as well as covariance matrix.
- Final layer of head is initialized to zero to start training with reasonable values.
- Extensive data augmentation is performed to avoid overfitting.



Learning Parameters from Data – Kalman Filter

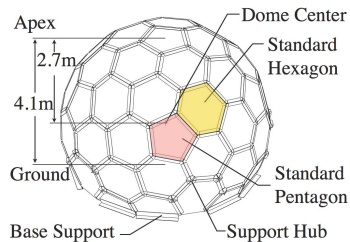
► Summary:

- Used only the annotations of the CMU Panoptic dataset.
- Dataset generated by extracting individual persons tracks from annotations.
- Learns all the parameters of the Kalman filter such as dynamics and constraints.
- Training by taking 100 frame segments of the ground truth tracks.
- Then generating synthetic observations, running filter, and minimizing MSE.



Dataset Overview – CMU Panoptic Dataset

- ▶ Source:
 - Video captured in a specialized 5.49m diameter dome-shaped room.
 - Equipped with hundreds of inward-facing cameras attached to the outside panels.
- ▶ Format and Data Characteristics:
 - 84 video sequences totaling approximately 11.5 hours of video.
 - VGA cameras recording at a 640×480 resolution at 25 fps.
 - Includes ground truth 3D skeletons and tracks in COCO19 format.



Experimental Setup

▶ Setup:

- Run evaluation on a subset of 8 sequences from the dataset.
- Compare generated pose tracks against ground truth annotations.
- Evaluate using varying number of camera views and differently sized YOLO models.
- Evaluate using different configurations of the tracking system.
- Also record timing statistics for evaluation of algorithm efficiency.

▶ Evaluation Metrics:

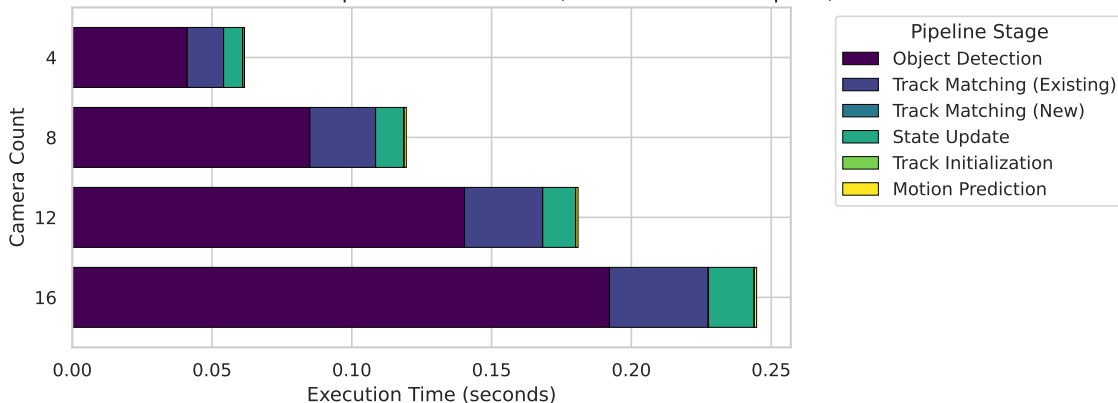
- Mean Per-Joint Position Error (MPJPE).
- Multi-Object Tracking Accuracy (MOTA).
- Also collected Precision and Recall.

Quantitative Results – Runtime Efficiency

► Main Finding:

- Detection is the largest part of the execution time per frame.
- Execution time of detection scales linearly with number of cameras.
- Matching and filtering also scale with camera count but slight less quickly.

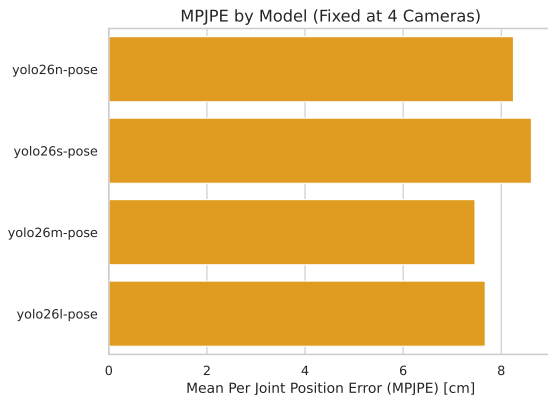
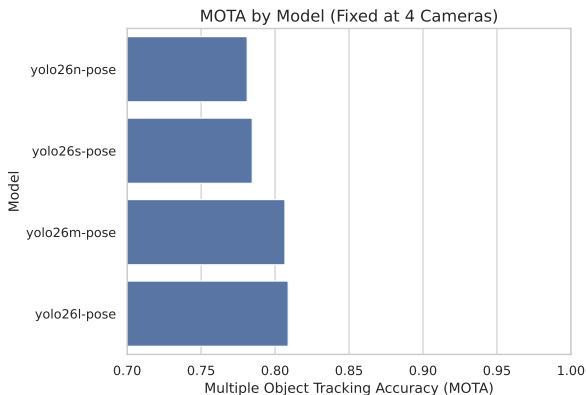
Execution Time Breakdown per Camera Count (Fixed at YOLO26n-pose)



Quantitative Results – Performance vs. Model Size

► Main Finding:

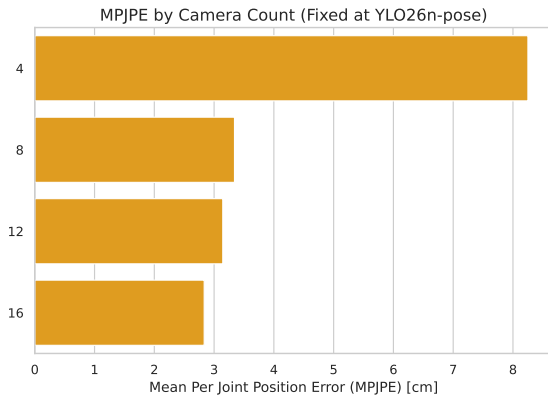
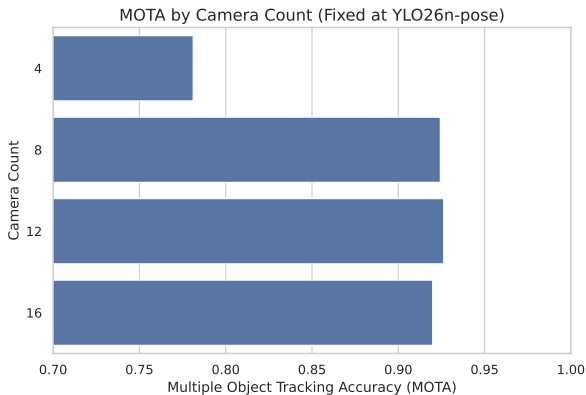
- Larger models only moderately improve performance.
- Largest performance jump is observed between the small and medium models.
- Looking at MPJPE there is no clear advantage for larger models.



Quantitative Results – Performance vs. Camera Count

- ▶ Main Finding:

- Adding more cameras significantly improves MPJPE.
- Additional views slightly reduce MOTA due to more false positives.
- More camera views is more effective than larger model capacity for MPJPE.

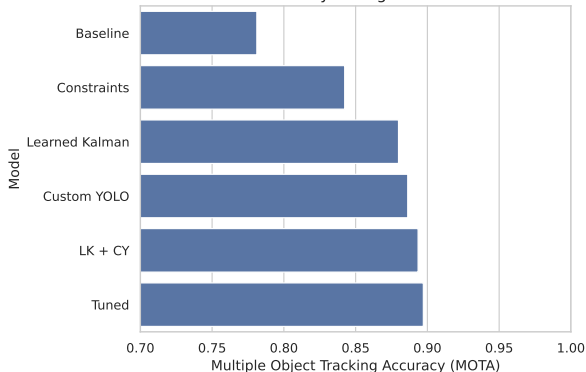


Quantitative Results – Skeletal Constraints and Learning

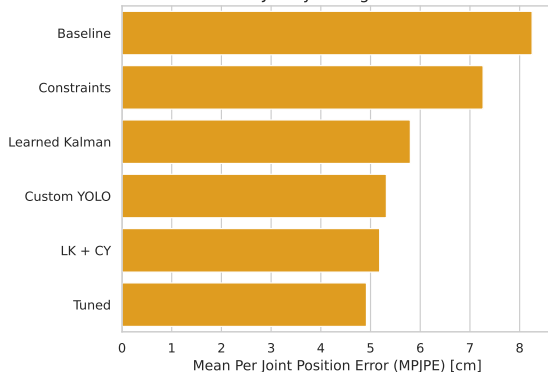
► Main Finding:

- Skeletal constraints give a significant boost in MOTA and MPJPE.
- Both learned Kalman and custom YOLO head improve performance.
- Tuning thresholds give a small improvement.

MOTA by Configurations

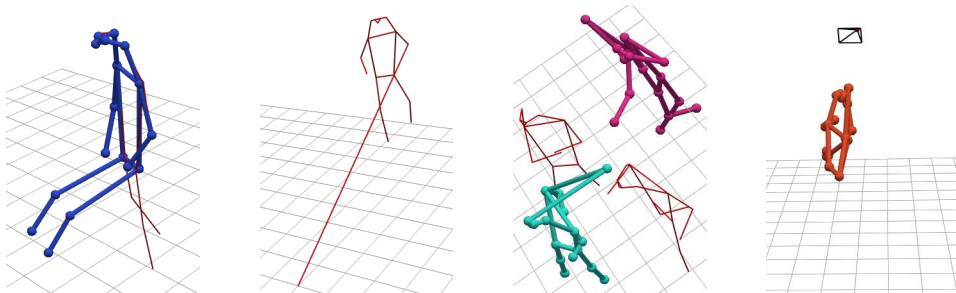


MPJPE by Configurations



Qualitative Results

- ▶ Common failure modes:
 - Partially occluded persons keypoints drifting.
 - Inactive track seen by only one camera can not cause track creation.
 - Active track seen by only one camera can cause drift.
 - Detection of person while exiting when ground truth is missing.



Conclusion

- ▶ **Summary:**

- Implemented an EKF framework that merges multi-view observations.
- Incorporated skeletal length consistency constraints.
- Implemented simple learning of parameters from data.
- Analyzed performance across different system configuration.

- ▶ **Limitations:**

- Still struggles significantly with occlusions.
- Running 2D pose estimation remains a performance bottleneck.

- ▶ **Possible Future Improvements:**

- Add more constraints that force other known properties.
- Find more stable learning methods for the Kalman filter parameters.
- Use of a particle filter with occlusion aware likelihood.

